

EX-10.22 5 fs42021a1ex10-22_amplitude.htm INTRAMURAL CLINICAL TRIAL AGREEMENT, EXECUTED FEBRUARY 11, 2021, BY AND BETWEEN JASPER THERAPEUTICS, INC. AND NATIONAL HEART, LUNG, AND BLOOD INSTITUTE, PART OF THE NATIONAL INSTITUTES OF HEALTH

Exhibit 10.22

*****Certain identified information has been omitted from this exhibit because it is both (i) not material and (ii) would likely cause competitive harm to the Registrant if publicly disclosed. Such omitted information is indicated by brackets (“...***...” in this exhibit. *****

IND SPONSOR: NHLBI

INTRAMURAL CLINICAL TRIAL AGREEMENT

For Clinical Trials Conducted at the National Institutes of Health Clinical Center

BETWEEN

National Heart, Lung, and Blood Institute

Part of the

National Institutes of Health

(“NIH”)

and

Jasper Therapeutics, Inc.

(hereinafter referred to as the “Company”)

titled

Phase I and II Study on the Addition of CD117 Antibody to Alemtuzumab, Low Dose Irradiation, and Sirolimus in Matched Related Donor Transplantation in Beta-Globin Disorders

The **National Heart, Lung, and Blood Institute**, an institute of the **NIH**, an agency of the U.S. Department of Health and Human Services

(hereafter, referred to as “**NHLBI**”),

on behalf of its employee, **Dr. John Tisdale**

(hereafter, referred to as the “**NHLBI Investigator**”),

and the **Company**, having an office at 2200 Bridge Parkway, Ste. 102, Redwood City, CA 94065 (each individually a “**Party**”, and collectively the “**Parties**”), have agreed to cooperate in the conduct of collaborative research related to a clinical trial to be held at the NIH Clinical Center, in Bethesda, Maryland, titled “**Phase I and II Study on the Addition of CD117 Antibody to Alemtuzumab, Low Dose Irradiation, and Sirolimus in Matched Related Donor Transplantation in Beta-Globin Disorders.**”

This Agreement sets forth the terms and conditions under which this protocol will be conducted and the clinical trial will be managed.

The Company and NHLBI agree as follows:

1. DEFINITIONS

The terms listed in this Section have the meanings indicated throughout this Agreement. To the extent a definition of a term as provided in this Section is inconsistent with a corresponding definition in the applicable sections of either the United States Code (U.S.C.) or the Code of Federal Regulations (C.F.R.), the definition in the U.S.C. or C.F.R. will control.

1.1. “**Adverse Event**” (**AE**) means any untoward medical occurrence in a Human Subject to whom the Test Article has been administered. An AE does not necessarily have a causal relationship with the Test Article, that is, it can be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of the Test Article, whether or not it is related to it. Adverse Event is further defined in ICH E6 section 1.2 and 21 C.F.R 312.32.

1.2. “**Affiliates**” With respect to the Company, “Affiliates” means,

- (i) any legal entity of which the securities or other ownership interests representing fifty per cent (50%) or more of the equity or fifty per cent (50%) or more of the ordinary voting power or fifty per cent (50%) or more of the general partnership interest are, at the time such determination is being made, owned, controlled or held, directly or indirectly, by such legal entity, or,
- (ii) any legal entity which, at the time such determination is being made, is controlling or under common control with, such legal entity.

As used herein, the term “control”, whether used as a noun or verb, refers to the possession, directly or indirectly, of the power to direct, or cause the direction of, the management or policies of a legal entity, whether through the ownership of voting securities, by contract or otherwise.

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- 1.3. **“Agreement”** means this Clinical Trial Agreement or “CTA”, all executed amendments and supplements to this Agreement and all schedules to this Agreement.
- 1.4. **“Case Report Form” (CRF)** means the data collection form(s) to be completed for each Human Subject participating in the Clinical Trial.
- 1.5. **“Certificate of Confidentiality” (CoC)** means a certificate issued by NIH pursuant to Section 301(d) of the Public Health Service Act (42 U.S.C. 241(d)), that protects the privacy of Human Subjects enrolled in the Protocol. With limited exceptions defined in 42 U.S.C. 241(d), the CoC protects from disclosure names or any information, documents or biospecimens containing ISI collected under a Protocol conducted under this CTA.
- 1.6. **“Clinical Research Site”** means the NIH Clinical Center, their contractors, and Investigators at Bethesda, Maryland, USA where the Clinical Trial will be conducted in strict accordance with the Protocol.
- 1.7. **“Clinical Study Report”** in accordance with ICH E6 Section 1.13, is a written description of a Clinical Trial in which the clinical and statistical description, presentations, and analyses are fully integrated into a single report. The Clinical Study Report contains information on results including reactogenicity, AEs, immunogenicity and other clinical or laboratory observations made with respect to the intervention employed in conducting the trial. A detailed description of the contents of a Clinical Trial Report is found in ICH E3 “Structure and Content of Clinical Study Reports.”
- 1.8. **“Clinical Trial”** is defined by the NIH as a research study in which one or more human subjects are prospectively assigned to one or more interventions (which may include placebo or other control) to evaluate the effects of those interventions on health-related biomedical or behavioral outcomes. Rules and regulations are codified at 45 CFR 46.102(d) and (f). In this Agreement, Clinical Trial means the Clinical Trial for the Protocol.
- 1.9. **“Completion of the Clinical Trial”** the Clinical Trial is complete when all data analyses under the Protocol have been performed and the Clinical Study Report has been submitted to the FDA.
- 1.10. **“Confidential Information”** means confidential scientific, proprietary, business, or financial information provided that Confidential Information does not include:
 - (a) information that is within the public domain prior to the time of the disclosure by the disclosing Party to the receiving Party or thereafter becomes within the public domain other than as a result of disclosure by the receiving Party or any of its representatives in violation of this Agreement;
 - (b) information that was, on or before the date of disclosure in the possession of the receiving Party without an obligation of confidentiality;
 - (c) information that is acquired by the receiving Party from a third party not under an obligation of confidentiality;

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[...***...]

- (d) information that is hereafter independently developed by the receiving Party, without reference to the Confidential Information received from the disclosing Party;
 - (e) information that the disclosing Party expressly authorizes the receiving Party to disclose.
 - (f) information that is reasonably required by scientific standards for publication of the results of the Clinical Trial (including Clinical Trial methods and/or data) or any information that is necessary for other researchers to verify the results of the Clinical Trial; or
 - (g) information that relates to potential hazards or cautionary warnings associated with the production, handling, or use of the Test Article.
- 1.11. **“Data and Safety Monitoring Board or Committee” (DSMB or DSMC)** is an independent group of experts that advises NHLBI and the Investigators. The primary responsibilities of the DSMB are to:
- (i) to periodically review and evaluate the accumulated data of the Clinical Trial for participant safety, Clinical Trial conduct and progress, and when appropriate, efficacy; and,
 - (ii) to make recommendations to NHLBI concerning the continuation, modification, or termination of the Clinical Trial.
- 1.12. **“Effective Date”** means the date of the last signature of the authorized representatives of the Parties executing this Agreement.
- 1.13. **“Food and Drug Administration” (FDA)** means the U.S. Food and Drug Administration.
- 1.14. **“Government”** means the federal government of the United States of America.
- 1.15. **“Human Subject”** means, in accordance with the definition in 45 C.F.R. § 46.102(f), a living individual about whom an Investigator conducting research obtains:
- (a) Data through intervention or interaction with the individual; or
 - (b) Identifiable Private Information.
- 1.16. **“Identifiable Private Information” (IPI)** means private information about a Human Subject from which the identity of the Human Subject is or may readily be ascertained. Regulations defining and governing this information include 45 C.F.R. Part 46 and 21 C.F.R. Part 50.
- 1.17. **“Identifiable, Sensitive Information” (ISI)** means, in accordance with the definition of 42 U.S.C. 241(d)(4), information that is about an individual and that is gathered or used during the course of research as described in 42 U.S.C. 241(d)(1)(A) through which an individual is identified, or that includes IPI, or for which there is at least a very small risk, as determined by current scientific practices or statistical methods, that some combination of the information, a request for the information, and other available data sources could be used to deduce the identity of an individual.

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- 1.18. **“Informed Consent Form”** means a signed and documented form in which each Human Subject voluntarily consents or confirms his or her willingness to participate in a particular clinical trial after having been informed of all aspects of the trial that are relevant to the subject’s decision to participate. The informed consent form satisfies the requirements of ICH E6, 45 C.F.R. Part 46 and 21 C.F.R. Part 50.
- 1.19. **“Institutional Review Board” (IRB)** means, in accordance with 45 C.F.R. Part 46, Protection of Human Subjects (Revised November 13, 2001) and 21 C.F.R. Part 56, Subpart C: IRB Functions and Operations (as amended June 18, 1991), and other applicable regulations, an independent body comprising medical, scientific, and nonscientific members, whose responsibility is to ensure the protection of the rights, safety, and well-being of the Human Subjects involved in a Clinical Trial. It may also be referred to as an Independent Ethics Committee in accordance with ICH E6, Section 1.27.
- 1.20. **“The International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use” (ICH)**. ICH refers to one or all of the following requirements used throughout this Agreement:
- (a) **ICH E2F**: “Development Safety Update Report”, including the latest finalized revision, if any.
 - (b) **ICH E3**: “Structure and Content of Clinical Study Reports”, including the latest finalized revision, if any.
 - (c) **ICH E6**: “Good Clinical Practice: Consolidated Guidance”, published in the Federal register (62 Federal Register 25692 (1997)), including the latest finalized revision. Also referred to as **“FDA Good Clinical Practice Guidelines”**.
 - (d) **ICH Q7**: “Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients” published in the Federal register (66 Federal Register 49028-9 (2001)), including the latest finalized revision, if any.
- 1.21. **“Invention”** means any invention or discovery that is or may be patentable or otherwise protected under Title 35 of the United States Code, or any novel variety of plant which is or may be protectable under the Plant Variety Protection Act, 7 U.S.C. §§ 2321 *et seq.*
- 1.22. **“Investigational New Drug Application” (IND)** is filed in accordance with 21 C.F.R. Part 312 under which clinical investigation of a Test Article (an experimental drug or biologic) is performed in Human Subjects in the United States or intended to support a United States licensing action.
- 1.23. **“Investigator”** means, in accordance with 21 C.F.R. § 312.3, an individual who actually conducts a clinical trial, that is, who directs the administration or dispensation of Test Article to a subject, and who assumes responsibility for studying Human Subjects, for recording and ensuring the integrity of research data, and for protecting the welfare and safety of Human Subjects. In this Agreement, “Investigator(s)” means the individual(s) identified as responsible for the conduct of the Clinical Trial at the Clinical Research Site.

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- 1.24. **“Investigator Brochure” (IB)** means, in accordance with the definition in 21 C.F.R. § 312.23(a)(5), a document containing information about the Test Article, including animal screening, preclinical toxicology, and detailed pharmaceutical data, including a description of possible risks and side effects to be anticipated on the basis of prior experience with the Test Article or related drugs, and precautions, such as additional monitoring, to be taken as part of the investigational use of the Test Article.
- 1.25. **“Office of Human Research Protections” (OHRP)** means the HHS office that oversees protection of human subjects from research risks under 45 C.F.R. Part 46 (the Common Rule).
- 1.26. **“Party”** means an entity entering into this Agreement, referred to individually as the “Party” and collectively as the “Parties”.
- 1.27. **“Patent”** means any issued United States patent, any international counterpart and any corresponding grant by a non-U.S. government in place of a patent.
- 1.28. **“Protocol”** means the formal, detailed description of the Clinical Trial to be performed as provided in the Protocol titled “Phase I and II Study on the Addition of CD117 Antibody to Alemtuzumab, Low Dose Irradiation, and Sirolimus in Matched Related Donor Transplantation in Beta-Globin Disorders.” A Protocol describes the objective(s), design, methodology, statistical considerations, and organization of a clinical trial. For the purposes of this Clinical Trial, the term Protocol includes any and all associated documents, including informed consent forms, to be provided to Human Subjects and potential participants in the Clinical Trial. This Clinical Trial Agreement will be governed by the most recent version of the Protocol, and should this Agreement be executed prior to complete finalization of the Protocol, the last-dated version thereof will be considered to be incorporated by reference in place of any prior versions. In the event that there is a conflict between the terms of the Protocol and the terms of the Agreement in other matters, the terms of this Agreement will govern.
- 1.29. **“Protocol Team”** means the team, under the direction of NHLBI, responsible for the development and management of the Protocol, evaluation of data, proposal of amendments, and all issues related to the Protocol or aspects of Protocol development and modification. The Protocol Team will include the representatives from the Company, if they wish to participate, the Principal Investigators, representatives from NHLBI, and the persons involved with statistical and data analysis for the Clinical Trial. Participation on the Protocol Team will be as agreed by the Parties and will take into account any special requirements of the Protocol design.
- 1.30. **“Serious AE” or “Serious SAR”** means an AE or SAR that in the view of either the Investigator or Sponsor, results in any of the following outcomes:
- (a) Death,
 - (b) A life-threatening AE,

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- (c) Inpatient hospitalization or prolongation of existing hospitalization,
- (d) A persistent or significant incapacity or substantial disruption of the ability to conduct normal life functions, or
- (e) A congenital anomaly/birth defect, or
- (f) Important medical events that may not result in death, be life-threatening, or require hospitalization, but, based upon appropriate medical judgment, they may jeopardize the patient or subject and may require medical or surgical intervention to prevent of the outcomes (a) through (e).

- 1.31. **“Sponsor”** means, in accordance with the definition in 21 C.F.R. § 312.3, an organization or individual who assumes legal responsibility for supervising or overseeing Clinical Trials with Test Article, and is sometimes referred to as the **“IND holder”**. The Sponsor for the Protocol is NHLBI.
- 1.32. **“Suspected Adverse Reaction”** or **“SAR”** means any AE for which there is a reasonable possibility that the drug caused the AE. For the purposes of IND safety reporting, “reasonable possibility” means there is evidence to suggest a causal relationship between the drug and the AE. Suspected Adverse Reaction is further defined in 21 C.F.R. § 312.32.
- 1.33. **“Test Article”** means, in accordance with 21 C.F.R. § 50.3(j), any drug (including a biological product), medical device, food additive, color additive, electronic product, material or any other article subject to regulation under the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. §§ 301, *et seq.*, Pub. L.No.75-717, 52 Stat. 1040 (1938), as amended. In this Agreement, JSP-191 is collectively referred to as the “Test Article”.
- 1.34. **“Unexpected Adverse Event”** or **“Unexpected Suspected Adverse Reaction”** means an AE or SAR which is considered “unexpected” because it is not listed in the Investigator Brochure or is not listed at the specificity or severity that has been observed; or, when an Investigator Brochure is not required or available, is not consistent with the risk information described in the general investigational plan or elsewhere in the current application. Unexpected Adverse Event and Unexpected Suspected Adverse Reaction are further defined in 21 C.F.R. § 312.32.

2. CLINICAL RESEARCH SITE AND INVESTIGATORS**2.1 NHLBI will not knowingly utilize:****2.1.1 Any organization performing services in connection with this Clinical Trial that has been:**

- (i) Debarred under the provisions of the Generic Drug Enforcement Act of 1992, 21 U.S.C. § 335a(a) and (b); or,
- (ii) Suspended by the Office for Human Research Protections (OHRP) as a clinical research site under 45 C.F.R. Part 46; and,

Jasper Therapeutics–NHLBI CTA

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2.1.2 Any person convicted of a felony under federal law for conduct:

- (i) Relating to the development or approval, including, but not limited to, the process for development or approval, of any drug, product, medical device, New Drug Application (NDA), Pre-Market Application (PMA), 510(k) or IND or similar application; or,
- (ii) Otherwise relating to the regulation of any drug product or medical device under the FD&C Act.
- (iii) Any person performing services in connection with this Clinical Trial that has been disqualified as a clinical investigator under 21 C.F.R. § 312.70.
- (iv) Any Investigator who is not qualified by training and experience as an appropriate expert to conduct the Clinical Trial, as required under 21 C.F.R. § 312.53.

2.2 If either Party becomes aware that any organization or person involved in the Clinical Trial is debarred, threatened with debarment, disqualified, threatened with disqualification, or suspended, that Party will notify the other Party immediately.

2.3 NHLBI will conduct the Clinical Trial in accordance with good clinical practice, including as defined by the ICH and comply with all applicable U.S. and foreign government, state and local laws, regulations and guidelines.

2.4 Company agrees that this Protocol will be conducted only at the Clinical Research Site. However, Company can conduct, at its own expense and under its own IND, additional Clinical Trials with the Test Article, at non-NHLBI-funded sites. Company agrees to inform NHLBI in writing of any other Clinical Trials it may support for the use of the Test Article that would compete with this Clinical Trial for the same Human Subject population. Company asserts that the Test Article constitutes proprietary technology.

3. INVESTIGATIONAL NEW DRUG APPLICATION SPONSORSHIP

3.1. **IND.** NHLBI will submit an IND covering the Protocol to the FDA. The IND will satisfy all of the requirements of the FDA. Company will provide a letter granting the FDA permission to cross-reference the Company's pertinent Drug Master File (DMF), New Drug Application (NDA), Biologics License Application (BLA), and/or INDs in support of the NHLBI for the limited purpose of the IND, and in return, NHLBI will also provide a letter to the Company, if requested, granting the FDA permission to cross-reference the IND filed by NHLBI for this Clinical Trial.

3.2. **Company holds only IND under which the Protocol(s) will be conducted.** During and for a period of [...***...] after the completion of a Protocol, the Company shall promptly provide to NHLBI any information that Company has reasonably determined could directly affect the health or safety of past or current Human Subjects or influence the conduct of the Protocol. Such information may arise from any source, for example, safety reports provided to the FDA, study results, information in site monitoring reports or data safety monitoring committee reports. NHLBI shall be free to communicate the relevant safety information to each Human Subject and the IRB.

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[...***...]

3.3. **Protocol Monitoring.** NHLBI will be responsible for Clinical Research Site monitoring and quality assurance of all data in accordance with the clinical monitoring plan. Monitoring will be done in compliance with FDA Good Clinical Practice Guidelines (ICH) (E6). NHLBI, subject to the restrictions in the **Certificate of Confidentiality Obligations** section, will communicate any clinically significant findings from clinical monitors to Company in a timely manner.

3.4. **Safety Reporting**

3.4.1. NHLBI will collect safety reports according to the procedure outlined in the Protocol. NHLBI will assume responsibility for the reporting of safety reports to the FDA and will provide copies of the reports to Company.

3.4.2.

(i) For any safety report that meets all of the following criteria of: (i) Serious (ii) Unexpected and (iii) Suspected Adverse Reaction, NHLBI will provide to Company a completed copy of the safety report at the time the report is submitted to the FDA, NHLBI will provide follow up information to Company at the time the follow up safety report is submitted to FDA. The reporting will be completed in the timeframes consistent with 21 CFR § 312.32.

(ii) NHLBI will report all other Serious and non-serious AEs to the FDA and to Company on a timely basis consistent with 21 C.F.R. § 312.33.

3.4.3. As the manufacturer, Company will, in a timely manner consistent with FDA requirements and during the term of this Clinical Trial, provide NHLBI with any information it now has or may obtain in the future regarding the safety and/or the toxicity of Test Article.

3.5. **Safety Monitoring.** In accordance with NIH guidelines, Company and NHLBI agree that the following type(s) of safety monitoring is (are) necessary and appropriate for this Clinical Trial:

Data and Safety Monitoring Board (DSMB). If a DSMB is constituted for the Clinical Trial, NHLBI will notify Company in advance of any DSMB review. Company may participate in and will receive the open session reports of the DSMB. The recommendations derived from the closed sessions will also be communicated to Company. Prior to the Completion of the Clinical Trial, data and reports distributed for DSMB review will be used only for the purposes of the DSMB meeting and will be held in confidence by the DMSB.

4. **FDA MEETINGS/COMMUNICATIONS**

4.1 With respect to any discussions with the FDA involving data obtained from this Clinical Trial under NHLBI's IND, NHLBI, in consultation with Company, will take the initiative in arranging meetings or conference calls with the FDA. Formal meetings with the FDA concerning the Clinical Trial design and/or data will be discussed and agreed upon in advance by Company and NHLBI. Company will have the right to participate in all formal meetings with the FDA. Company agrees not to contact the FDA independent of NHLBI concerning this Clinical Trial. However, Company may contact the FDA on separate product-related issues.

Jasper Therapeutics–NHLBI CTA

[...***...]

- 4.2 Company will promptly notify NHLBI of any FDA correspondence related to the Protocol that is received by Company, or its Affiliates; any FDA enforcement actions directed toward Company or its Affiliates, including but not limited to: warning letters, seizures, recalls, injunctions/consent decrees, rejection of regulatory submissions or withdrawal of approval for a Test Article, criminal investigations, and proceedings to debar Company or its Affiliates or individuals employed under a contract to Company and/or its Affiliates.
- 4.3 Company will also promptly notify NHLBI of any action taken by the FDA regarding manufacturing of the Test Article that would impact the safety of Human Subjects in the Clinical Trial.

5. SUPPLY, DISTRIBUTION, AND USE OF TEST ARTICLE**5.1. Supply.**

NHLBI will provide Company with an estimate of the quantity of Test Article that will be required to complete the Protocol.

- 5.1.1 Company will supply the Test Article to NHLBI [...***...] and in quantities and conditions sufficient to complete the Protocol and on a schedule mutually agreed upon by the Parties to ensure a sufficient supply of unexpired Test Article.
- 5.1.2 Company will be responsible for labeling the Test Article used in the Clinical Trial.

5.2. Distribution.

- 5.2.1 Company will ship the Test Article to the Clinical Research Site as mutually agreed by the Parties. Company will provide specific storage and/or shipping instructions for the Test Article to NHLBI, who will be responsible for adhering to them, as mutually agreed by the Parties. Company warrants that any packaging for hazardous material provided by Company meets Department of Transportation regulatory requirements for use at the Clinical Research Site.
- 5.2.2 The Test Article must be received by the Clinical Research Site in usable condition and accompanied by Material Safety Data Sheet (MSDS), specific storage and shipping instructions, stability and/or expiration dating information and the finally signed and dated Certificate of Analysis (COA) for each lot of Test Article sent. If Company performs ongoing stability testing for each lot of Test Article sent, then Company will also provide updated retest or expiration dates for those respective lots to NHLBI in a timely manner.
- 5.2.3 If there is evidence that the Test Article that arrived at the Clinical Research Site has not been maintained according to the defined shipping instructions or there is evidence of damage to the Test Article container or container closure system, NHLBI will contact Company to inform them of the condition of the received Test Article and to determine together with the Company whether the Test Article is usable or if it must be replaced. During the course of the Clinical Trial, the same process will be used whenever there is evidence that the Test Article has not been maintained according to Company's recommended storage conditions. If the Test Article must be replaced, Company will replace it at [...***...] to NHLBI.

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[...***...]

5.3. Use.

5.3.1 NHLBI will neither transfer the Test Article to any third party nor will NHLBI chemically modify, replicate, make derivatives of, or reverse engineer the Test Article unless required by the Protocol or mutually agreed in writing by the Parties.

5.3.2 NHLBI will request that the Investigator:

- (i) use the Test Article only in accordance with the Protocol and for no other purpose;
- (ii) not transfer the Test Article to any parties except Company; and
- (iii) not chemically modify, replicate, make derivatives of, or reverse engineer the Test Article unless required by the Protocol or as mutually agreed to, in writing, by the Parties.

5.4. **Investigator Brochure.** Company will provide a current Investigator Brochure for all applicable components of the Test Article, and any later revisions and addenda to the Investigator Brochure for the Test Article to NHLBI, as mutually agreed by the Parties, who will agree to keep them in confidence in accordance with Section 11 (Confidential Information) of this Agreement.

5.5. **Warranty.** Company represents and warrants that the Test Article supplied shall be manufactured and released according to the principles of current Good Manufacturing Practice and when administered in accordance with Protocol it is suitable for human use.

5.6. **Source.** In the event Company elects to terminate its development of Test Article for reasons other than safety, without the transfer of its development efforts and obligations under this Agreement to another party acceptable to NHLBI within [...***...] of discontinuation, then Company will provide NHLBI with Test Article for all then-enrolled Human Subjects sufficient to complete the Clinical Trial in the manner described in the Protocol.

5.7. **Termination of Development.** Company hereby grants to NHLBI a nonexclusive, nontransferable, irrevocable, paid-up license to practice or have practiced for or on behalf of the United States any invention which Company may have or obtain on Test Article, its manufacture, or on the process for use of Test Article, throughout the world, for medical research purposes related to the Protocol for the sole purpose of completing the Clinical Trial. This license will only become effective in the event Company terminates its development of Test Article for reasons other than safety or IRB approval withdrawal, without the transfer of its development efforts to another party within [...***...] of termination, and NHLBI elects to continue the development of Test Article. This provision will become null and void upon FDA approval of the Test Article indications and marketing of the Test Article by Company.

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[...***...]

6. PROTOCOL DEVELOPMENT

- 6.1 The Parties agree that enrollment in the Clinical Trial will not start until the version of the Protocol to be used has been reviewed in advance by the Protocol Team; approved (stipulations met/resolved) by the relevant IRB(s) and NHLBI in writing; and submitted to the FDA, the thirty (30) calendar day wait period has been satisfied and any FDA clinical hold issues have been responded to satisfactorily. The Protocol is a product of NHLBI and will be deemed NHLBI Confidential Information, as defined in Section 11 (Confidential Information) of this Agreement.
- 6.2 The Parties agree that any alteration in or amendment to the Protocol must be accepted by the Protocol Team, and approved in writing by the relevant IRB(s) and NHLBI and submitted to the FDA, if appropriate, prior to such alteration or amendment becoming effective. Company will review any amendments or alterations to the Protocol and NHLBI will consider such comments in good faith in finalizing such amendments or alterations.

7. CASE REPORT FORM DEVELOPMENT

NHLBI or its designee will be responsible for the development and subsequent revisions, if any, of the Case Report Forms with appropriate review and comment by the Protocol Team.

8. HUMAN SUBJECTS PROTECTION

- 8.1 NHLBI and Company recognize the principles of respect for persons, beneficence (including minimization of harms and maximization of benefits), and justice as stated in the Belmont Report and will apply these principles in all research covered under this Agreement. The informed consent of each Human Subject participating in the Clinical Trial at the Clinical Research Site will be obtained prospectively using an IRB approved informed consent process. The informed consent document may be reviewed in advance by Company and will be approved by NHLBI and all appropriate Institutional Review Board (IRB). NHLBI will consider in good faith any comments provided by Company.
- 8.2 NHLBI and Company acknowledge and accept their responsibilities for protecting the rights and welfare of human research subjects set forth in 45 C.F.R. Part 46, Protection of Human Subjects (Revised January 19, 2017).

Therefore:

- 8.2.1. NHLBI will maintain the confidentiality of IPI of Human Subjects collected under the Clinical Trial and protect the privacy of each of the individual Human Subjects in the Clinical Trial unless disclosure is required by law. Company will not have access to any IPI of Human Subjects collected under the Clinical Trial.
- 8.2.2. NHLBI may inspect, but not copy, Human Subjects' medical records that might also include information not directly connected to this Clinical Trial. However, NHLBI agrees that this information will remain confidential and will not be used for any purpose other than confirmation of Clinical Trial data.

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- 8.2.3. NHLBI agrees not to include IPI that could lead to identification of individual Human Subjects in any release of data, reports or publications related to the Clinical Trial.
- 8.2.4. NHLBI agrees not to use IPI about Human Subjects for any purpose not stated in the Protocol without the consent of Company and local site IRB approval.
- 8.2.5. NHLBI and Company agree to comply with the determinations of all IRBs overseeing this research.
- 8.2.6. Human Subject specimens or data provided to Company during or after the Clinical Trial will be coded. Unequivocally, neither IPI nor the key linking coded data or specimens to Human Subjects will be released to Company.

9. DATA ANALYSIS AND MANAGEMENT, CLINICAL SPECIMENS AND ISOLATES

- 9.1 NHLBI will be responsible for the data, and scientific reporting of all results/data obtained from the Clinical Trial at the Clinical Research Site.
- 9.2 NHLBI will have responsibility for the data management: collection, entry, and quality control edits (with implied verifications and documentation) and analysis of data obtained from the Clinical Trial in accordance with the Protocol.
- 9.3 All data obtained from the Clinical Trial will be in the custody of NHLBI. However, NHLBI will not use the data obtained from the Clinical Trial for purposes of seeking regulatory approval without first obtaining the prior written permission of the Company.
- 9.4 Upon completion of the data analyses, NHLBI will authorize transfer to Company a copy of the complete data analysis set in a machine-readable format to be determined jointly by the Parties. If Company requires that the data be provided to it in any customized format(s), Company will [...***...] associated with the customized data format(s). When applicable, compressed raw and intermediate genomics and/or other -omics data will be provided in industry/scientific community accepted data formats (e.g. FASTQ and/or BAM format for RNA-Sequence transcriptomics studies) at the time of data generation or transfer. Light-weight derived data such as gene expression values per sample will be provided in SAS format.
- 9.5 NHLBI is responsible for sending a final Clinical Study Report to Company and the FDA as applicable, according to specified content within a reasonable time after the dataset is locked.
- 9.6 Subject to the right of NHLBI to publish the data from this Clinical Trial as set forth in Section 10 (Publications and Press Releases) of this Agreement, Company has the right to utilize the data reports in its possession and all de-identified data obtained from this Clinical Trial for all legitimate business or regulatory purposes, and NHLBI agrees to promptly provide all such data to Company in accordance with Section 9.4. NHLBI and Company may provide any information regarding the Clinical Trial to governmental organizations including, but not limited to, the FDA, and the Securities and Exchange Commission (SEC), for all legitimate public health, regulatory, or business purposes. Except for information related to regulatory or safety issues or under emergency circumstances where it is not practicable to do so and to the extent permitted by law, NHLBI will not release information regarding the Clinical Trial to governmental organizations without prior notification of the release information to the Company.

Jasper Therapeutics–NHLBI CTA

[...***...]

- 9.7 For applicable clinical trials, NHLBI will post applicable primary and/or secondary endpoint results within [...***...] of the last visit for primary and/or secondary end point evaluations. Therefore, data that will be provided by the Company for primary or secondary outcomes (i.e. immunogenicity analysis) will be provided to NHLBI no later than [...***...] after the last visit of the last patient evaluated for the primary endpoint.
- 9.8 **Clinical Specimens for Use in Protocol.** All clinical specimens and isolates collected under this Clinical Trial will be and remain in the custody of NHLBI. However, NHLBI will provide Company with clinical specimens and isolates for analysis as stated in the Protocol. Company agrees to share with NHLBI results derived from the analysis of clinical specimens and isolates.
- 9.9 **Clinical Specimens for Use in Future Research.** At the Company's request, and with the written approval of the appropriate NHLBI authorizing official, NHLBI will, only when available and as permitted by the Informed Consent Form, provide Company with some aliquots of clinical specimens, identified only by Clinical Trial subject number and accompanied by related encoded data for future studies. As stated in Section 8 (Human Subjects Protection) of this Agreement, NHLBI will provide to Company only encoded data and encoded samples and will not provide the link between the code and the Human Subjects. For clarification purposes, NHLBI will not provide any IPI to Company. Nothing in this Agreement will be interpreted to transfer ownership to Company.
- 9.10 Upon completion of the studies involving clinical specimens or upon the termination of this Agreement, whichever comes first, any unused clinical specimens and data derived from Human Subjects in the Company's possession will be destroyed unless NHLBI gives Company directions for disposing of the clinical specimens and data by another means.

10. PUBLICATIONS and PRESS RELEASES

- 10.1 Any publications based on the results of the Clinical Trial and originating from NHLBI will conform to NIH publication policies. Unless requested otherwise by Company, NHLBI will acknowledge Company as the source of the Test Article in any NHLBI publication resulting from the Clinical Trial.
- 10.2 Recognizing that employees of either Party may play an important role in the design, analysis, and interpretation of the findings of the Clinical Trial, each Party will include appropriate individuals from the other Party in the authorship of publications resulting from the Clinical Trial, in accordance with the generally accepted customs pertaining to authorship.

Each Party will provide the other Party with a copy of any abstract, scientific meeting presentation, or manuscript prior to submission for publication with sufficient time [**Abstracts:** [...***...]; **Manuscripts:** [...***...]] for review and comment]. Each Party agrees that, following the receiving Party's review of the abstract and/or manuscript for the maximum periods of time specified above, if no comment is received by the submitting Party, the submitting Party will be free to publish, present or use any Clinical Trial data. The receiving Party will maintain the proposed publication or public disclosure of the submitting Party as Confidential Information until publication or public disclosure by the submitting Party.

Jasper Therapeutics–NHLBI CTA

[...***...]

- 10.3 If Company requests additional time to file a patent application related to the publication: The publication or other disclosure may be delayed for up to [...***...], upon written request by either Party as necessary to preserve U.S. or foreign patent or other intellectual property rights.
- 10.4 Each Party will provide a copy of any proposed press release to the other Party for review and comment at least [...***...] in advance of proposed publication. Each Party agrees that, following the receiving Party's review of the proposed press release for the maximum periods of time specified above, if no comment is received by the submitting Party, the submitting Party will be free to publish the press release.

11. CONFIDENTIAL INFORMATION

- 11.1 **Certificate of Confidentiality Obligations.** Any ISI that Company receives from NHLBI is covered by a CoC and therefore all copies of ISI are immune from the legal process, and will not, without the consent of the Human Subject, be admissible as evidence or used for any purpose in any action, suit, or other judicial, legislative, or administrative proceeding.

Notwithstanding the foregoing, Company will be permitted to disclose ISI:

- (a) If required by Federal, State, or local laws (e.g., as required by the Federal Food, Drug, and Cosmetic Act, or state laws requiring the reporting of communicable diseases to State and local health departments), excluding instances of disclosure in any Federal, State, or local civil, criminal, administrative, legislative, or other proceeding;
 - (b) If the consent of the Human Subject to whom the information, document, or biospecimen pertains obtained by NHLBI allowed for such disclosure.
 - (c) Prior to making any permitted disclosures above, Company will ensure that any recipient of ISI protected by a CoC is aware of its confidential nature and the requirement to comply with the CoC (see <https://humansubjects.nih.gov/coc/background>).
- 11.2 Either Party may disclose and/or receive Confidential Information under the terms and conditions of this Agreement. Each receiving Party will limit its disclosure and use of the disclosing Party's Confidential Information to the amount necessary to conduct the Clinical trial, and will place a confidentiality notice on all the Confidential Information. The disclosing Party will reduce confidential non-written communications to writing within [...***...] of first disclosure. Each Party receiving Confidential Information agrees that any information so designated will be maintained as Confidential Information in accordance with this Agreement and used by it only for the purposes of the Clinical Trial. Any Party may object to the designation of information as Confidential by the other Party.

Jasper Therapeutics–NHLBI CTA

[...***...]

- 11.3 Unless expressly provided otherwise, neither Party will disclose, copy, reproduce or otherwise make the disclosing Party's Confidential Information available to any other person or entity without the consent of the disclosing Party unless required by a court or administrative body of competent jurisdiction, the Freedom of Information Act (FOIA), 5 U.S.C. § 552, 45 C.F.R. Part 5, or other applicable laws and/or regulations to disclose the Confidential Information, except that NHLBI may disclose Company's Confidential Information to other parties, as required for the conduct of the Clinical Trial for purposes of the Clinical Trial. NHLBI will request such parties receiving Company's Confidential Information to maintain the confidentiality of Confidential Information consistent with the terms of this Agreement.
- 11.4 Each Party will use the same level of care it uses with its own Confidential Information, but no less than a reasonable level of care, in maintaining the confidentiality of the other Party's Confidential Information. While NHLBI will endeavor to control the distribution of the Protocol document itself, Company acknowledges that some Government documents are available (with abstracts) to the public under the Freedom of Information Act. In addition, NHLBI requires the posting of information on the [ClinicalTrials.gov](https://clinicaltrials.gov) registry of clinical studies, available through the NIH Website, consistent with the Food and Drug Administration Amendments Act of 2007, 121 STAT. 823.
- 11.5 Each Party agrees that the receiving Party is not liable for the disclosure of Confidential Information which, after notice to and consultation with the disclosing Party, the receiving Party determines may not be lawfully withheld, provided the disclosing Party has been given an opportunity to seek a court order to enjoin from disclosure.
- 11.6 Each Party's obligation to maintain the confidentiality of Confidential Information will expire at the earlier of the date when the information is no longer Confidential Information as defined above or [...***...] after the expiration or termination date of this Agreement. Either Party may request an extension to this term when necessary to protect Confidential Information relating to [...***...].

12. INTELLECTUAL PROPERTY

- 12.1 Ownership of any Invention conceived solely or jointly by NHLBI as a consequence of conducting the Clinical Trial and involving the Test Article, will be determined under U.S. laws pertaining to intellectual property created in the course of federally funded research. Neither Party acquires by virtue of this Agreement any right, title, nor interest in or to any issued Patents or pending patent applications owned or controlled by the other Party. Nothing in this Agreement will be construed as granting any license or obligation to license any intellectual property owned by Company to NHLBI with respect to the Test Article other than the limited right to use the Test Article for the performance of the Protocol in accordance with the terms of this Agreement.
- 12.2 **NHLBI Intellectual Property.**
- 12.2.1 The Government will retain title to any Patent, pending patent applications or other intellectual property rights in Inventions conceived solely by NHLBI employees in the course of the clinical research.

Jasper Therapeutics–NHLBI CTA

[...***...]

12.2.2 NHLBI agrees to notify Company of any NHLBI sole or joint Invention and to disclose it to Company under an appropriate confidentiality agreement. Company may apply for nonexclusive or exclusive license rights to any such patentable Invention made by NHLBI employees that might arise during the clinical research and NIH will consider in good faith Company's application for a nonexclusive or exclusive license consistent with 37 C.F.R. Part 404. NHLBI does not have the authority to grant present rights in future NHLBI-owned inventions, however, granting internal-use research licenses is within the mission of NHLBI. NHLBI agrees that if the Parties enter a Cooperative Research and Development Agreement (CRADA) that relates to the subject matter of this Agreement, it will include terms to be negotiated then that provide to the Company (i) a non-exclusive, royalty-free, internal-use research license, and (ii) a first and exclusive option for Company to negotiate for a royalty-bearing exclusive license (or, at Company's election, non-exclusive license) to any sole or joint Invention for commercial use, in all cases subject to and consistent with 37 C.F.R. Part 404.

12.3 **Company Intellectual Property.** Company will retain title to any Patents, pending patent applications, or other intellectual property rights in Inventions conceived by its employees during the course of the clinical research.

12.4 **Joint NHLBI-Company Intellectual Property.** NHLBI and Company will have joint intellectual property rights in Inventions conceived jointly by their employees during the course of the clinical research. Company may apply for an exclusive license to NHLBI rights in such Inventions in accordance with the 37 CFR 404, which application will be considered in good faith by NHLBI.

13. **FORCE MAJEURE**

Neither Party will be liable for any unforeseeable event beyond its reasonable control not caused by the fault or negligence of such Party, which causes such Party to be unable to perform its obligations under this Agreement, and which it has been unable to overcome by the exercise of due diligence. In the event of the occurrence of such a force majeure event, the Party unable to perform will promptly notify the other Party. It will further use its best efforts to resume performance as quickly as possible and will suspend performance only for such period of time as is necessary as a result of the force majeure event.

14. **LIABILITY, INDEMNIFICATION, INSURANCE & RESEARCH RELATED INJURY**

14.1 **Liability.** In view of the Anti-Deficiency Act, 31 U.S.C § 1341, NHLBI cannot agree to indemnify Company for its losses. Each Party will be liable for the losses, claims, damages, or liabilities that it incurs as a result of its activities under this Agreement except that NHLBI, as an agency of the Government, assumes liability only to the extent provided under the Federal Tort Claims Act, 28 U.S.C. Ch. 171.

14.2 **Indemnification.** Company will defend, indemnify and hold harmless NHLBI, and its grantees and contractors and their respective agents and employees ("Indemnitee(s)") from any and all liabilities, damages, losses, claims, action, suits and expenses, including attorneys' fees and court costs (collectively "Claims") to the extent caused by the administration or use of the Test Article in the Clinical Trial. Company's control over the defense and settlement of any claim against NHLBI will be subject to the consent of NHLBI and the Department of Justice. The Indemnitee(s) will at all times have the right to fully participate in the defense of any Claim at their own expense and for their own account. Company's obligation to so indemnify Indemnitee(s) will only apply if each of the following conditions is met:

14.2.1 The Claim was not proximately caused by the Indemnitee(s)' failure to conduct the Clinical Trial in accordance with the Protocol and this Agreement;

Jasper Therapeutics–NHLBI CTA

[...***...]

- 14.2.2 The Claim was not caused by the negligence, recklessness, willful misconduct of any Indemnitee or violation of law or regulation or breach of this Agreement by NHLBI, provided that any action properly taken by the Indemnitee in compliance with the Protocol or written instructions from the Company will be deemed, for purposes of this condition, not to be negligent, and provided further that if a Claim is jointly caused by the negligence of any Indemnitee and the administration or use of the Test Article, then Company will provide defense and indemnification to the extent the Claim was caused by the administration of the Test Article;
- 14.2.3 The Claim was not caused by any alteration in or amendment to the Protocol that was not approved in writing by Company;
- 14.2.4 The Company is promptly notified of the Claim;
- 14.2.5 The Company will have the ability to participate jointly with the Department of Justice in the defense and settlement of the claim, subject to the requirements of 28 U.S.C. 516, 518, and 519;
- 14.2.6 The Company will have the right to select its own defense counsel to undertake the joint defense and settlement of the Claim.
- 14.3 The Company will provide a diligent defense against and/or settlement of any Claims for which defense and indemnification are provided under this Agreement whether such claims are rightfully or wrongfully made. The Company will have the right to settle such Claims, at the Company's sole expense and in the Company's sole discretion. Indemnitee(s) will fully cooperate, at the Company's expense, with the Company and its legal representatives in the investigation and defense of any Claim for which defense and indemnification are provided under this Agreement.
- 14.4 **Insurance.** Company represents and warrants that it will maintain during the term of this Agreement or the Protocol, whichever is longer, a liability insurance policy or a program of insurance or self-insurance at levels sufficient to support the indemnification obligations assumed under this Agreement and to cover the costs of medical care required to treat or stabilize adverse reactions suffered by Human Subjects who received Test Article in accordance with the approved Protocol. Upon request, Company will provide evidence of its insurance or self-insurance to NHLBI.
- 14.5 **Human Subject Injury or Illness Attributable to the Test Article.** Company will pay the reasonable cost of medical care required by Human Subjects for illness or injury attributable to the Test Article, to the extent such costs are (i) not covered by the Human Subject's medical or hospital insurance or by third party or governmental programs providing such coverage, and (ii) not the result of the Human Subject's pre-existing abnormal medical condition or underlying disease to the extent that the Human Subject failed to disclose such pre-existing abnormal medical condition or underlying disease and the pre-existing medical condition or underlying disease was part of the exclusion criteria, or gross negligence by NHLBI. For purposes of this determination and Company's obligation under this Agreement, "attributable" means that the receipt of the Test Article and the Clinical Trial Human Subject's illness or injury are reasonably related in time, and the illness or injury is more likely explained by the receipt of the Test Article than any other cause. The payment or offer of payment of any amount by Company on behalf of a Human Subject or his or her healthcare insurer or other third party payer under this Section is not an admission of fault or liability by any one or more of (a) the Government or any agency thereof; or (b) Company, its employees or agents, and any such payment or offer of payment will not be considered a waiver of any defense or other legal right by any of the foregoing in any legal, administrative or similar proceeding.

Jasper Therapeutics–NHLBI CTA

[...***...]

14.6 **Limitation on Liability.** In no event shall Company be liable to NHLBI or any third party for any special incidental or consequential damages arising out of or relating to this Agreement, or the subject matter hereof, however caused and whether such claim is based in contract, tort (including negligence and strict liability) or otherwise, even if authorized representative of Company is advised of the possibility of such damages. Except for Company's indemnification obligations hereunder, the total, cumulative liability of Company arising out of or related to this Agreement shall not exceed [...***...] U.S. dollars (\$[...***...]).

14.7 **Disclaimer.** Except as specifically stated in Section 5.5 or in representations or warranties made elsewhere in this Agreement, the Test Article is provided "AS IS" and "WITH ALL FAULTS", Company expressly disclaims any warranties, express or implied, of any kind, including any implied warranties of merchantability, fitness for a particular purpose or noninfringement of intellectual property rights of third parties.

15. DISPUTES

Any dispute arising under this Agreement that is not disposed of by Agreement of the Parties will be submitted jointly to the signatories of this Agreement. If the signatories are unable to jointly resolve the dispute within [...***...] after notification thereof, the dispute will be referred to the Director of NHLBI (or his/her designee) and an appropriate authorized representative of Company for resolution. If the Director of NHLBI (or his/her designee) and the authorized representative of Company are unable to jointly resolve the dispute within [...***...], either Party may pursue any and all administrative or judicial remedies that may be available.

16. INDEPENDENT CONTRACTORS

In the performance of all work under this Agreement, neither Party is authorized or empowered to act as agent for the other for any purpose and will not, on behalf of the other Party, enter into any contract, warranty, or representation as to any matter. Neither Party will be bound by the acts of the other Party.

17. NON-ENDORSEMENT

By entering into this Agreement, NHLBI does not directly or indirectly endorse any product or service provided, or to be provided, by Company. Company will not in any way state or imply that this Agreement is an endorsement of those product(s) or service(s) by the Government or any of its organizational units or employees. However, Company may reference or use publications and reports based on the Clinical Trial for legitimate business and regulatory purposes.

Jasper Therapeutics–NHLBI CTA

[...***...]

18. AMENDMENTS

Modifications to this Agreement will not be effective unless made in writing, as mutually agreed, and signed by a duly authorized representative of each Party.

19. SURVIVABILITY

The provisions of Sections 2 (Clinical Research Site and Investigators), 3 (Investigational New Drug Application Sponsorship), 4 (FDA Meetings/Communications), 5 (Supply, Distribution, and Use of Test Article), 8 (Human Subjects Protection), 9 (Data Analysis and Management, Clinical Specimens and Isolates), 10 (Publications and Press Releases), 11 (Confidential Information), 12 (Intellectual Property), 14 (Liability, Indemnification, Insurance and Research Related Injury), 17 (Non-Endorsement) and this Section 19 (Survivability); will survive the expiration or earlier termination of this Agreement.

20. ENTIRE AGREEMENT AND SEVERABILITY

This Agreement constitutes the entire Agreement and understanding of the Parties with respect to the subject matter hereof and supersedes any prior understanding or written or oral Agreement. The provisions of this Agreement are severable and, in the event that any provision of this Agreement will be determined to be invalid or unenforceable under any controlling body of law, such determination will not in any way affect the validity and enforceability of the remaining provisions of this Agreement.

21. ASSIGNMENT

Neither this Agreement nor any rights or obligations of any Party hereunder will be assigned or otherwise transferred by either Party without the prior written permission of the other Party, provided, however, that Company, with prior written notice to NHLBI, may assign this Agreement at any time without the consent of NHLBI to any Affiliate of Company or to any third party in connection with a sale of all or substantially all of the business or assets of Company that relate to this Agreement to such third party, provided that the assignee commits to continuing the supply of the Test Article until completion of the Clinical Trial.

22. APPLICABLE LAW

This Agreement will be construed in accordance with Federal law as applied by the Federal courts of the District of Columbia.

23. TERM AND TERMINATION

23.1 Unless terminated sooner in accordance with this Section 23, this Agreement will expire upon receipt of the final study report by Company.

Intramural Clinical Trial Agreement

Document Printed February 4, 2021

Page 20 of 22

Jasper Therapeutics–NHLBI CTA

[...***...]

- 23.2 The Parties may terminate this Agreement at any time by mutual written consent.
- 23.3 Either Party may unilaterally terminate this Agreement at any time by giving written notice at least thirty (30) calendar days prior to the desired termination date.
- 23.4 The Parties may agree that this Agreement should be terminated prior to completion of the Protocol. However, the Protocol will be completed if medically or ethically appropriate. In that event, each enrolled Human Subject will be followed through the period outlined in the Protocol and Company will supply enough Test Article to complete the Protocol.
- 23.5 In the event Company elects to terminate its obligations under the terms of this Agreement, due to an unexpected dissolution, Company must notify NHLBI within at least thirty (30) calendar days of the dissolution; and, provide NHLBI with the resources necessary to complete the Protocol in addition to the terms of Section 5 above.

24. NOTICES

Any notice or report required under the terms of this Agreement will be sent to the other Party at the addresses indicated below. Any notice will be deemed to be effective when delivered to the other Party by courier, registered mail (with return receipt), via facsimile, Portable Document Format (PDF), or email followed by conformational hard copies when requested.

For NHLBI:

Office of Technology Transfer and Development
 For legal matters: National Heart, Lung, and Blood Institute
 National Institutes of Health
 Building 31, Room 4A29, MSC 2479
 31 Center Drive
 Bethesda, MD 20892-2479

For technical/clinical John F. Tisdale, M.D.
 matters: Cellular and Molecular Therapeutics Branch
 National Heart, Lung, and Blood Institute
 National Institutes of Health
 [...***...]
 10 Center Drive
 Bethesda, MD 20814
 [...***...]

For Company:

[...***...]
 For legal matters: Jasper Therapeutics, Inc.
 2200 Bridge Parkway, Ste. 102
 Redwood City, CA 94065
 [...***...]

For technical/clinical [...***...]
 matters: Jasper Therapeutics, Inc.
 2200 Bridge Parkway, Ste. 102
 Redwood City, CA 94065

SIGNATURES BEGIN ON THE NEXT PAGE

Jasper Therapeutics–NHLBI CTA

[...***...]

This Agreement may be executed in counterparts, each of which shall be an original as against either Party whose signature appears thereon, but all of which taken together shall constitute but one and the same instrument. An executed facsimile or electronic scanned copy of this Agreement shall have the same force and effect as an original.

If Company agrees with the terms of this Agreement for the Clinical Trial in accordance with the Protocol entitled “Phase I and II Study on the Addition of CD117 Antibody to Alemtuzumab, Low Dose Irradiation, and Sirolimus in Matched Related Donor Transplantation in Beta-Globin Disorders”, please have an authorized representative sign below.

FOR NHLBI:

Duly Authorized:/s/ Richard Childs**Signature of NHLBI Authorizing Official #1**2/11/2021**Date**Richard Childs. M.D.**Name of NHLBI Authorizing Official #1**Clinical Director. DIR. NHLBI**Title of NHLBI Authorizing Official #1**/s/ Bruce D. Goldstein -S**Signature of NHLBI Authorizing Official #2**02.05.2021**Date**Bruce Goldstein, Esq.**Name of NHLBI Authorizing Official #2**Director, Office of Technology Transfer and Development,
NHLBI**Title of NHLBI Authorizing Official #2**

FOR COMPANY:

Duly Authorized:/s/ William Lis**Signature of Company's Authorizing Official**2/11/2021**Date****Intramural Clinical Trial Agreement**

Page 22 of 22

*Read and Understood by the NHLBI Investigator:***I hereby acknowledge that I have read and understand the terms of this Agreement.**/s/ John Tisdale__ John Tisdale, MD**Signature of NHLBI Investigator**02/05/2021**Date**Bill Lis**Name of Company's Authorizing Official**Chief Executive Officer**Title of Company's Authorizing Official**

Document Printed February 4, 2021

